Indeterminate Forms and L'Hopital's Rule

Section 3.7

Suppose that $f(x)$ and $g(x)$ are differentiable, $g'(x) \neq 0$ and that $\lim\limits_{x\to a} \dfrac{f(x)}{g(x)} = \dfrac{0}{0}$ or that $\lim\limits_{x\to a} \dfrac{f(x)}{g(x)} = \pm\dfrac{\infty}{\infty}$ (i.e., we have an in determinant form of the type $\dfrac{0}{0}$ or $\dfrac{\infty}{\infty}$), then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$$

A. Indeterminate form $\frac{\infty}{\infty}$ or $\frac{0}{0}$

If we have a limit of the form $\lim\limits_{x\to a} \dfrac{f(x)}{g(x)}$ where both $f(x) \to 0$ and $g(x) \to 0$, then we have the in determinant form of type $\dfrac{0}{0}$

If we have a limit of the form $\lim\limits_{x\to a} \dfrac{f(x)}{g(x)}$ where both $f(x) \to \infty$ and $g(x) \to \infty$, then we have the in determinant form of type $\dfrac{\infty}{\infty}$.

Examples

1. **(video)** $\lim\limits_{x\to 2} \dfrac{2x^2 - 8}{x - 2}$

2. **(video)** $\lim\limits_{x\to 0} \dfrac{3e^{2x} - 3}{x}$

3. **(video)** $\lim_{x \to 1} \frac{3-3x}{5\ln x}$

4. **(video)** $\lim_{x \to \infty} \frac{x^2+x+3}{2x^2+5}$

5. **(video)** $\lim_{x \to \infty} \frac{x+3}{\sqrt{x}+1}$

6. $\lim_{x \to -2} \frac{x^2-4}{x+2}$

7. $\lim_{x \to 0} \frac{1-e^{ax}}{x^7}$, assume $a > 0$

8. $\lim_{x \to 0} \frac{\sin(10x)}{\sin(bx)}$

9. $\lim_{x\to\infty} \frac{\ln x}{x^2}$

10. $\lim_{x\to 1} \frac{\ln x}{x-1}$

11. $\lim_{x\to 0} \frac{\sin x}{1-\cos x}$

12. $\lim_{x\to 0} \frac{a^x - 5^x}{9x}$

13. $\lim_{x\to 0} \frac{1+x-e^x}{3x^2}$

B. Indeterminate form $\infty \cdot 0$

Hint: Remember, in order to use L'Hopital's Rule the expression must be in the form $\lim_{x\to a} \frac{f(x)}{g(x)}$

If not, our first step is to get it in that form.

If $\lim_{x\to a} f(x)\cdot g(x)$ equals either $0\cdot\infty$ or $\infty\cdot 0$ then you will need to rewrite it first.

You could either rewrite it as $\lim_{x\to a} \frac{f(x)}{\left(\frac{1}{g(x)}\right)}$ or $\lim_{x\to a} \frac{g(x)}{\left(\frac{1}{f(x)}\right)}$

*Try putting the EASY function on the bottom!

Examples

14. $\lim_{x\to 0^+} x^4 \ln x$

15. $\lim_{x\to 0} \cot(2x)\cdot\sin(6x)$

16. $\lim_{x\to\infty} x^5 e^{-x^4}$

C. Other Indeterminate Forms

1. $\infty - \infty$
2. 0^0
3. ∞^0
4. 1^∞

Examples of the form "$\infty - \infty$"

17. $\lim\limits_{x \to 0^+} \csc(ax) - \cot(ax)$

Examples of the form "0^0, ∞^0, or 1^∞"

18. $\lim\limits_{x \to 0^+} x^x$

19. $\lim_{x\to\infty}\left(1+\frac{2}{x}\right)^{3x} =$

20. $\lim_{x\to 0^+}(1-7x)^{\frac{1}{x}} =$